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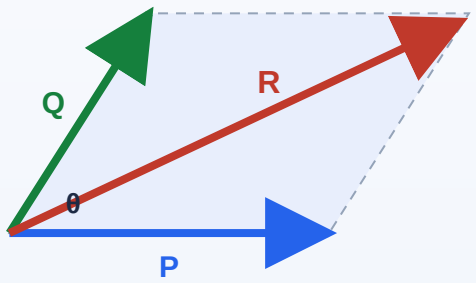
UPSSSC JE

MECHANICAL ENGINEERING MECHANICS

Complete English Edition

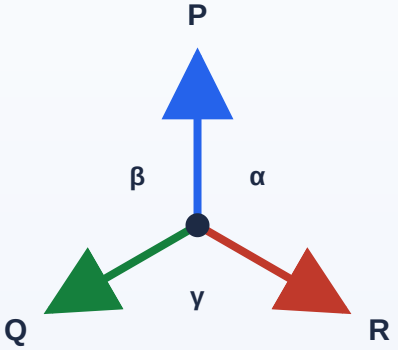
Unit 2 · 10 Marks · 2026

Parallelogram Law



$R = \sqrt{(P^2 + Q^2 + 2PQ \cos \theta)}$

Lami's Theorem



$P/\sin \alpha = Q/\sin \beta = R/\sin \gamma$

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UPSSSC JE (Mechanical) 2026: Part-1 has **10 questions (10 marks)** from **Engineering Mechanics**. Syllabus: equilibrium of forces, laws of motion and friction. This PDF follows every point of the official syllabus (Advertisement No. 24-Exam/2026).

Each chapter contains key points, diagrams, a formula table, detailed explanation, solved numericals, one-liners, MCQs with explanations and SSC JE previous year questions. Content follows standard BTEUP diploma textbooks (R. S. Khurmi, R. K. Bansal).

For a better score, study this PDF and also attempt the QUIZ.

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Chapter 1: Equilibrium of Forces

Equilibrium of Forces – Basic Concept

Force is an external agent that changes or tends to change the state of rest or motion of a body. Force is a **vector quantity**; its magnitude, direction, point of application and line of action must all be known. When the resultant of all forces acting on a body is zero and there is no turning effect, the body is said to be in **equilibrium**.

Understand in Detail — Equilibrium of Forces

1.1 What is a force — and why is it called a vector?

To open a door we push it, to lift a bucket we pull it — this push or pull is **force**. A force is an external agent that changes or tends to change the state of rest or motion of a body. SI unit **newton (N)** — 1 N is the force that produces an acceleration of 1 m/s² in a mass of 1 kg. 1 kgf = 9.81 N.

To describe a force completely four things are needed: **magnitude** (how much), **direction** (which way), **nature** (push or pull) and **point of application** (where). Since force has both magnitude and direction, it is a **vector** quantity — so two forces cannot be added like ordinary numbers. Forces of 3 N and 4 N at right angles give a resultant not of 7 N but of 5 N!

Force systems: collinear — all forces on one line (tug of war); **concurrent** — lines of action of all forces meet at a point (several ropes on one hook); **parallel** — lines of action parallel (loads on a beam); **general (non-concurrent, non-parallel)** — forces on a ladder. **Coplanar** — all forces in one plane.

Principle of transmissibility: the point of application of a force on a rigid body can be moved anywhere along its line of action without changing its effect — push a cart from behind or pull it from the front along the same line, the result is the same.

1.2 Resultant force — parallelogram law and the method of components

The **resultant** is the single force that alone produces the combined effect of all the forces.

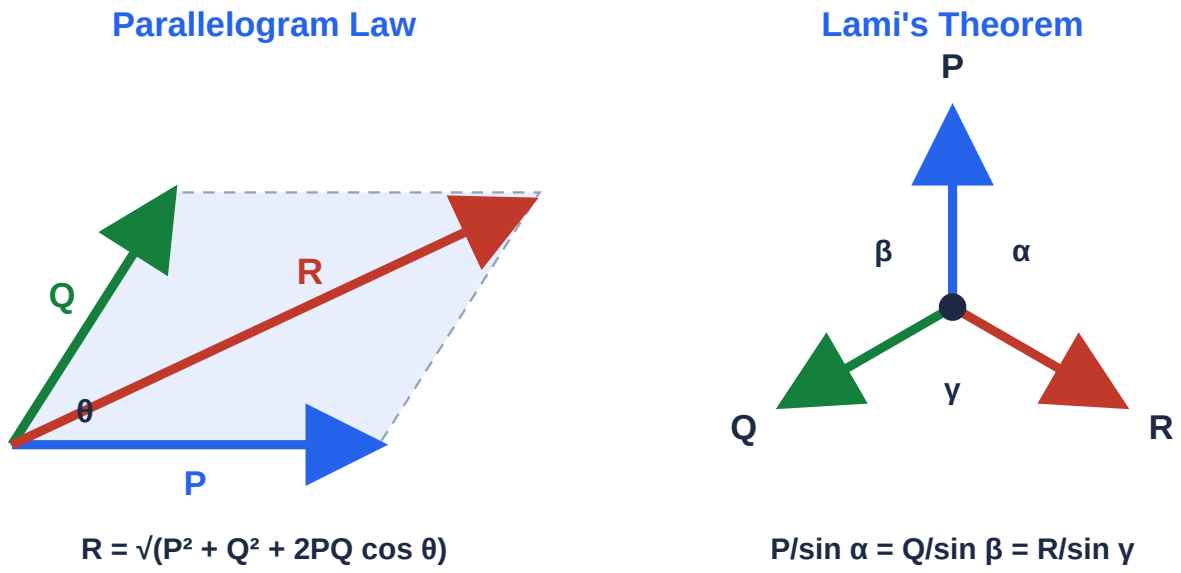


Fig 2.1: Parallelogram law of forces and Lami's theorem

Derivation: Parallelogram law of forces

1. Two forces P and Q at angle θ from a point — represent them by two adjacent sides of a parallelogram ; the diagonal = resultant
2. **$R = \sqrt{(P^2 + Q^2 + 2PQ\cos\theta)}$**
3. Angle with P: **$\tan\alpha = Q\sin\theta/(P + Q\cos\theta)$**
4. Special cases: θ = 0° → R = P + Q (maximum) ; θ = 180° → R = P − Q (minimum) ; θ = 90° → R = $\sqrt{(P^2 + Q^2)}$; if P = Q then R = 2P cos(θ/2)

Derivation: Method of components — for many forces

1. Break each force F into horizontal (F cos θ) and vertical (F sin θ) components — θ measured from horizontal
2. ΣH = sum of all horizontal components ; ΣV = sum of all vertical components (watch the signs!)
3. **$R = \sqrt{(\Sigma H^2 + \Sigma V^2)}$** ; **$\tan\alpha = \Sigma V/\Sigma H$** (α from horizontal) ; the signs of ΣH and ΣV fix the quadrant

Example: At a point act 20 N towards east, 30 N north and 10 N west. Resultant?

Solution:

1. ΣH = 20 − 10 = 10 N (east) ; ΣV = 30 N (north)
2. R = $\sqrt{(10^2 + 30^2)}$ = $\sqrt{1000}$ = **31.6 N**
3. tan α = 30/10 = 3 → α = **71.6°** from east towards north

1.3 Equilibrium — when nothing moves

A body is in **equilibrium** when the resultant of all forces on it is zero — it neither slides (no translation) nor turns (no rotation). **Equilibrant** = a force equal in magnitude and exactly opposite to the resultant, which brings the system into equilibrium.

Conditions of equilibrium (coplanar): ΣH = 0, ΣV = 0, ΣM = 0 — the first two prevent sliding, the third prevents turning. For concurrent forces the first two are enough (all pass through one point, so no turning).

Free body diagram (FBD) — the first step of every problem: draw the body 'separated' from its surroundings and show **all** forces acting on it — weight (down), tension (along the rope, away from the body), normal reaction (perpendicular to the surface), friction (opposing motion). Draw the FBD right and half the problem is solved.

Derivation: Lami's theorem

1. If **three concurrent forces** keep a point in equilibrium
2. then each force is proportional to the sine of the angle between the other two
3. **P/sin α = Q/sin β = R/sin γ** (α = angle between Q and R, etc.)
4. Condition: only three forces, concurrent, and exactly in equilibrium ; take the angle **opposite** to the force

Types of equilibrium: stable — returns when slightly disturbed (ball in a bowl); **unstable** — moves further away when disturbed (ball on an inverted bowl); **neutral** — stays wherever placed (ball on a flat surface).

Example: A 100 N weight hangs by two strings making 30° and 60° with the ceiling (horizontal). Tensions in the strings?

Solution:

1. Angle between strings = 180 − 30 − 60 = 90° ; between T_1 (30° string) and the weight = 90 + 30 = 120° ; between T_2 (60° string) and the weight = 90 + 60 = 150°
2. Lami (angle **opposite** each force): $T_1/\sin150° = T_2/\sin120° = 100/\sin90°$
3. $T_1 = 100 \times 0.5 =$ **50 N** ; $T_2 = 100 \times 0.866 =$ **86.6 N** — the steeper (60°) string carries more load ✔

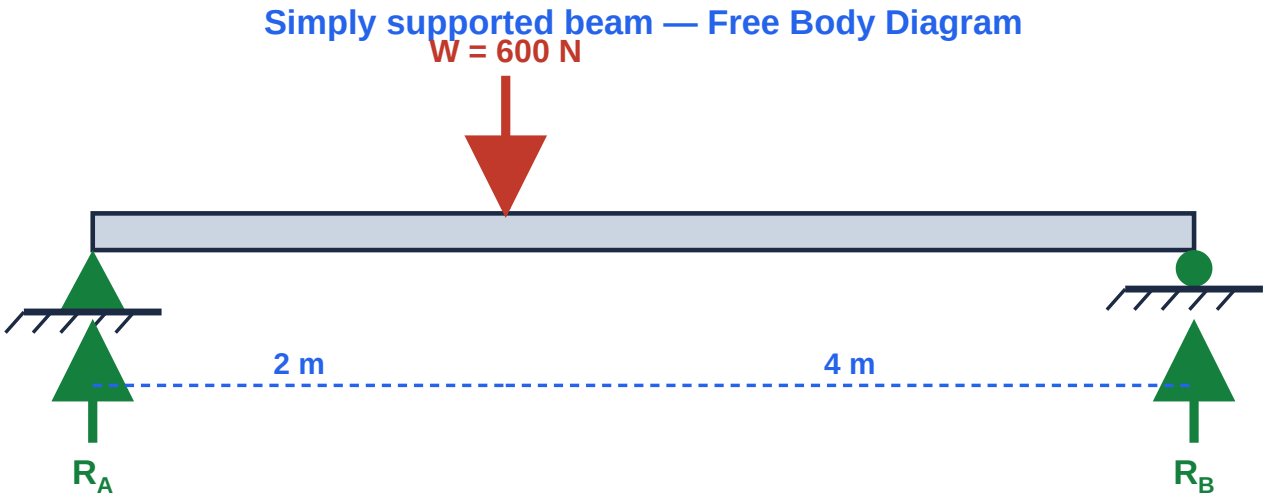
1.4 Moment, couple and Varignon's theorem

Push a door near its hinge and it needs great effort, push at the handle and it opens easily — because the turning effect depends not only on force but also on **distance**. **Moment = force × perpendicular distance from the point to the line of action**, M = F × d, unit N·m. Anticlockwise is usually taken positive.

Varignon's theorem (principle of moments): the algebraic sum of moments of several forces about a point = the moment of their resultant about the same point. It is used to find the **position** of the resultant: $\bar{x} = \Sigma M/R$.

Couple: two equal, opposite and parallel (but not collinear) forces — like turning a water tap or a car's steering wheel. Its resultant force is **zero**, so the body does not slide, it only **rotates**. Moment of couple = F × a (a = perpendicular distance between the forces, the **arm**) — it is the same about **any point**. A couple can be balanced only by another couple, not by a single force.

Centre of gravity and centroid: centre of gravity (CG) — the point where the whole weight of a body may be taken to act; **centroid** — the geometric centre of a plane figure. Standard: rectangle — centre; triangle — h/3 from base; semicircle — 4r/3π from diameter; solid hemisphere — 3r/8; hollow hemisphere — r/2; cone — h/4 from base.



$\Sigma M_A = 0 \rightarrow R_B \times 6 = 600 \times 2 \rightarrow R_B = 200\ N,\ R_A = 400\ N$

Fig 2.2: Support reactions by principle of moments

Example: On a 4 m beam, 20 kN acts at 1 m and 40 kN at 3 m from the left end A. Resultant and its position?

Solution:

1. R = 20 + 40 = **60 kN**
2. Moments about A: 20 × 1 + 40 × 3 = 140 kN·m
3. Varignon: 60 × $\bar{x}$ = 140 → $\bar{x}$ = **2.33 m** (from A) ; support reactions: R_B = 140/4 = 35 kN, R_A = 25 kN

1. Key Points

- **Unit of force:** newton (N) in SI; 1 N = 1 kg·m/s². 1 kgf = 9.81 N and 1 dyne = 10^{-5} N.
- **Types of force systems:** collinear (same line of action), concurrent (passing through one point), parallel, and non-concurrent non-parallel.
- **Parallelogram law of forces:** $R = \sqrt{(P^2 + Q^2 + 2PQ\cos\theta)}$ and $\tan\alpha = Q\sin\theta/(P + Q\cos\theta)$. At θ = 0°, R = P + Q (maximum); at θ = 180°, R = P − Q (minimum).
- **Conditions of equilibrium (coplanar forces):** ΣFx = 0, ΣFy = 0 and ΣM = 0. For concurrent forces only the first two are needed.
- **Lami's theorem:** if three concurrent forces are in equilibrium, each force is proportional to the sine of the angle between the other two: $P/\sin\alpha = Q/\sin\beta = R/\sin\gamma$.
- **Moment of a force:** M = F × perpendicular distance. Unit N·m. Clockwise and anticlockwise moments are taken with opposite signs.

- **Varignon's theorem:** the algebraic sum of moments of forces about a point equals the moment of their resultant about the same point.
- **Couple:** two equal, opposite, parallel forces with different lines of action. Moment of couple = F × arm; it is the same about every point and its resultant force is zero.
- **Free body diagram (FBD):** a sketch of the body isolated from its supports, showing all forces and reactions acting on it.
- **Polygon law of forces:** if several concurrent forces are represented in order by the sides of a polygon, the closing side taken in the opposite order gives the resultant.
- **Resultant of parallel forces:** for like parallel forces the resultant = sum; for unlike forces = difference. Its position is found by Varignon's theorem.
- **Equilibrant:** a force equal, opposite and collinear to the resultant; it brings the system into equilibrium.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Resultant of two forces	**$R = \sqrt{(P^2 + Q^2 + 2PQ\cos\theta)}$**
Direction of resultant	**$\tan\alpha = Q\sin\theta / (P + Q\cos\theta)$**
Forces at right angle ($\theta = 90°$)	**$R = \sqrt{(P^2 + Q^2)}$**
Resolution of a force	**$Fx = F\cos\theta$, $Fy = F\sin\theta$**
Resultant of several forces	**$R = \sqrt{[(\Sigma Fx)^2 + (\Sigma Fy)^2]}$, $\tan\alpha = \Sigma Fy/\Sigma Fx$**
Lami's theorem	**$P/\sin\alpha = Q/\sin\beta = R/\sin\gamma$**
Moment	**$M = F \times d$**
Moment of a couple	**$M = F \times a$ (a = arm)**

3. Detailed Explanation

- **Principle of transmissibility:** the point of application of a force on a rigid body may be moved anywhere along its line of action without changing its external effect.
- **Triangle law of forces:** if three forces represented in order by the sides of a triangle form a closed triangle, they are in equilibrium. The polygon law is its extension.
- **Equilibrium of two forces:** two forces are in equilibrium only if they are equal, opposite and collinear. For three forces in equilibrium they must be concurrent or parallel.
- **Supports and reactions:** a roller support gives one reaction normal to the surface; a hinge gives two components; a fixed support gives two forces and a moment.
- **Effect of a couple:** a couple produces only rotation, not translation. Shifting a force to another point gives a force plus a couple.
- **Replacing a force by a force and a couple:** moving a force F to a point O gives the same force F at O plus a couple of moment F × d. This is used in offset loads and bracket calculations.

- **Static determinacy:** a coplanar system has three independent equilibrium equations; if unknown reactions exceed three, the structure is statically indeterminate.
- **Forces in pin-jointed trusses:** members carry only axial (tension/compression) forces; in the method of joints ΣFx = ΣFy = 0 is applied at each joint.

4. Comparison Table — Force vs Couple

Basis	Force	Couple
Effect	Translation (and rotation)	Rotation only
Resultant force	Not zero	Zero
Moment	Depends on the point	Same about all points
Balanced by	An equal opposite force	Only an opposite couple

5. Solved Numerical Problems

Example 1. Two forces of 30 N and 40 N act at a point at 90°. Find the magnitude and direction of the resultant.

Solution:

1. $R = \sqrt{(30^2 + 40^2)} = \sqrt{2500} =$ **50 N**
2. $\tan \alpha = 40/30 = 1.333$, so $\alpha =$ **53.13°** (from the 30 N force)

Example 2. Two equal forces P act at 120°. Find the resultant.

Solution:

1. $R = \sqrt{(P^2 + P^2 + 2P^2 \cos 120°)} = \sqrt{(2P^2 - P^2)}$
2. R = **P** (resultant equals each force)

Example 3. A 6 m simply supported beam carries 600 N at 2 m from end A. Find the reactions.

Solution:

1. Moments about A: $R_B \times 6 = 600 \times 2$, so $R_B =$ **200 N**
2. $R_A = 600 - 200 =$ **400 N**

Example 4. A 100 N weight hangs from two strings inclined at 30° and 60° to the vertical. Find the tensions.

Solution:

1. Angle between strings = 90°; between T_1 (30° string) and weight 150°, between T_2 and weight 120°
2. Lami: $T_1/\sin 120° = T_2/\sin 150° = 100/\sin 90°$
3. $T_1 = 100 \times 0.866 =$ **86.6 N**, $T_2 = 100 \times 0.5 =$ **50 N**

Example 5. Three forces 10 N (0°), 20 N (90°) and 10 N (180°) act at a point. Find the resultant.
Solution:

1. $\Sigma Fx = 10 - 10 = 0$, $\Sigma Fy = 20$
2. R = **20 N**, direction vertical (90°)

Example 6. A couple with arm 2 m has moment 100 N·m. Find the force.
Solution:

1. F = M/a = 100/2 = **50 N**

6. Frequently Asked Facts

- Lami's theorem applies only to three concurrent forces in equilibrium.
- A couple has zero resultant force but a non-zero moment.
- The resultant of two equal forces equals each force when the angle between them is 120°.

7. Note (Confusing Facts)

- Force and moment are vectors, but work and energy are scalars.
- At $\theta = 90°$ the resultant is $\sqrt{(P^2 + Q^2)}$, not P + Q.
- A couple cannot be balanced by a single force.

Remember: One line: for equilibrium $\Sigma Fx = \Sigma Fy = \Sigma M = 0$. Assertion–Reason: (A) A couple only rotates a body. (R) Its resultant force is zero. — Both true; R explains A.

30 Most Important One-Liner Facts — Equilibrium of Forces

1. SI unit of force?

— Newton (N).

2. 1 kgf equals?

— 9.81 N.

3. Force is which type of quantity?

— Vector.

4. Forces on the same line of action?

— Collinear forces.

5. Forces passing through one point?

— Concurrent forces.

6. Maximum resultant of two forces?

— P + Q ($\theta = 0°$).

7. Minimum resultant of two forces?

— P − Q ($\theta = 180°$).

8. Resultant of two forces at 90°?

— $\sqrt{(P^2 + Q^2)}$.

9. Lami's theorem applies to how many forces?

— Three concurrent forces.

10. Unit of moment?

— N·m.

11. Principle of moments is due to?

— Varignon.

12. Resultant force of a couple?

— Zero.

13. Effect of a couple?

— Rotation only.

14. Principle of transmissibility?

— Force may be moved along its line of action.

15. Reactions at a roller support?

— One (normal to surface).

16. Reactions at a hinge?

— Two components.

17. Reactions at a fixed support?

— Two forces + one moment.

18. When are two forces in equilibrium?

— Equal, opposite and collinear.

19. A closed force triangle means?

— Forces are in equilibrium.

20. Resolution of a force?

— F cos θ and F sin θ.

21. Equilibrant?

— Force equal and opposite to resultant.

22. Polygon law is for?

— Several concurrent forces.

23. Resultant of like parallel forces?

— Sum.

24. Resultant of unlike parallel forces?

— Difference.

25. Equilibrium equations in a plane?

— Three.

26. Truss members carry?

— Only axial forces.

27. Moving a force to another point gives?

— Force + couple.

28. Moment is which quantity?

— Vector.

29. Usual sign of clockwise moment?

— Negative.

30. FBD stands for?

— Free Body Diagram.

Practice MCQs — Equilibrium of Forces

Q1. The resultant of two equal forces equals either force. The angle between them is:

(a) 60°
(b) 90°
(c) 120°
(d) 180°

✔ **Answer: (c) 120°**

Explanation: $R^2 = 2P^2(1 + \cos\theta) = P^2$ gives $\cos\theta = -½$, $\theta = 120°$.

Q2. Lami's theorem applies to:

(a) Two forces
(b) Three concurrent forces
(c) Parallel forces
(d) Any number of forces

✔ **Answer: (b) Three concurrent forces**

Explanation: Only three concurrent forces in equilibrium.

Q3. The resultant force of a couple is:

(a) Zero
(b) Twice the force
(c) Equal to the force
(d) Infinite

✔ **Answer: (a) Zero**

Explanation: The two forces are equal and opposite.

Q4. Resultant of 30 N and 40 N at right angles is:

(a) 70 N
(b) 10 N
(c) 50 N
(d) 35 N

✔ **Answer: (c) 50 N**

Explanation: √(900 + 1600) = 50 N.

Q5. The principle of moments is known as:

(a) Lami's
(b) Varignon's
(c) Newton's
(d) D'Alembert's

✔ **Answer: (b) Varignon's**

Explanation: Varignon's theorem.

Q6. Number of unknown reactions at a fixed support:

(a) 1
(b) 2
(c) 3
(d) 4

✔ **Answer: (c) 3**

Explanation: Two force components and one moment.

Q7. Conditions needed for equilibrium of concurrent forces:

(a) ΣFx = 0
(b) ΣFy = 0
(c) ΣFx = 0 and ΣFy = 0
(d) ΣM = 0

✔ **Answer: (c) ΣFx = 0 and ΣFy = 0**

Explanation: For concurrent forces the moment is automatically zero.

Q8. Moving a force along its line of action does not change its effect — this is:

(a) Principle of transmissibility
(b) Principle of superposition
(c) Lami's theorem
(d) Triangle law

✔ **Answer: (a) Principle of transmissibility**

Explanation: Principle of transmissibility.

Q9. The resultant of two forces is maximum when the angle between them is:

(a) 0°
(b) 90°
(c) 120°
(d) 180°

✔ **Answer: (a) 0°**

Explanation: At θ = 0°, cos θ = 1 and R = P + Q.

Q10. 1 kgf equals:

(a) 1 N
(b) 9.81 N
(c) 10^5 dyne
(d) 0.981 N

✔ **Answer: (b) 9.81 N**

Explanation: 1 kgf = 1 kg × 9.81 m/s².

Q11. The equilibrant is ____ the resultant:

(a) Equal and in the same direction
(b) Equal, opposite and collinear
(c) Half
(d) Double

✔ **Answer: (b) Equal, opposite and collinear**

Explanation: It cancels the resultant.

Q12. Independent equations for equilibrium of coplanar non-concurrent forces:

(a) 1
(b) 2
(c) 3
(d) 6

✔ **Answer: (c) 3**

Explanation: ΣFx, ΣFy, ΣM.

Q13. Members of a truss carry:

(a) Bending moment
(b) Only axial forces
(c) Torsion
(d) Shear force

✔ **Answer: (b) Only axial forces**

Explanation: Loads act at pin joints.

Q14. 10 N and 10 N at 60° — resultant:

(a) 10 N
(b) 14.1 N
(c) 17.3 N
(d) 20 N

✔ **Answer: (c) 17.3 N**

Explanation: √(100 + 100 + 200 × 0.5) = √300 = 17.3.

Q15. When a force is shifted to another point, we must add:

(a) A force
(b) A couple
(c) Friction
(d) Nothing

✔ **Answer: (b) A couple**

Explanation: A couple of F × d.

SSC JE / UPSSSC JE Previous Year Questions — Equilibrium of Forces

Q1. If three concurrent forces are in equilibrium, each is proportional to the ____ of the angle between the other two: SSC JE (Previous Year)

(a) sine
(b) cosine
(c) tangent
(d) cotangent

✔ **Answer: (a) sine**

Explanation: Lami's theorem.

Q2. Moment of a couple equals: SSC JE (Previous Year)

(a) Force × arm
(b) Force ÷ arm
(c) Force × velocity
(d) Zero

✔ **Answer: (a) Force × arm**

Explanation: M = F × a.

Q3. Two forces of 5 N and 12 N act at right angles. The resultant is: SSC JE (Previous Year)

(a) 17 N
(b) 7 N
(c) 13 N
(d) 60 N

✔ **Answer: (c) 13 N**

Explanation: √(25 + 144) = 13 N.

Q4. Reaction at a hinged support is: SSC JE (Previous Year)

(a) Only vertical
(b) Only horizontal
(c) Both vertical and horizontal
(d) A moment

✔ **Answer: (c) Both vertical and horizontal**

Explanation: A hinge gives two components, no moment.

Q5. By the parallelogram law, resultant of two forces: SSC JE (Previous Year)

(a) $\sqrt{(P^2 + Q^2 + 2PQ \cos \theta)}$
(b) $\sqrt{(P^2 + Q^2 - 2PQ \cos \theta)}$
(c) P + Q
(d) $\sqrt{(P^2 - Q^2)}$

✔ **Answer: (a) $\sqrt{(P^2 + Q^2 + 2PQ \cos \theta)}$**

Explanation: Standard formula.

Q6. Sum of moments of forces about a point equals moment of the resultant — this is: SSC JE (Previous Year)

(a) Lami's theorem
(b) Varignon's theorem
(c) Newton's law
(d) Hooke's law

✔ **Answer: (b) Varignon's theorem**

Explanation: Varignon.

Chapter 2: Laws of Motion

Laws of Motion – Basic Concept

Dynamics studies the motion of bodies together with the forces causing it. Newton's three laws of motion form the foundation of mechanics, and quantities such as momentum, impulse, work, power and energy are built on them.

Understand in Detail — Laws of Motion

2.1 Describing motion — displacement, velocity, acceleration

Distance — total length of path travelled (scalar); **displacement** — straight-line distance from start to end with direction (vector). After one round of a circular ground, distance = circumference but displacement = zero. **Speed** = distance/time (scalar); **velocity** = displacement/time (vector). **Acceleration** = rate of change of velocity, a = dv/dt, unit m/s²; decreasing velocity = **retardation**.

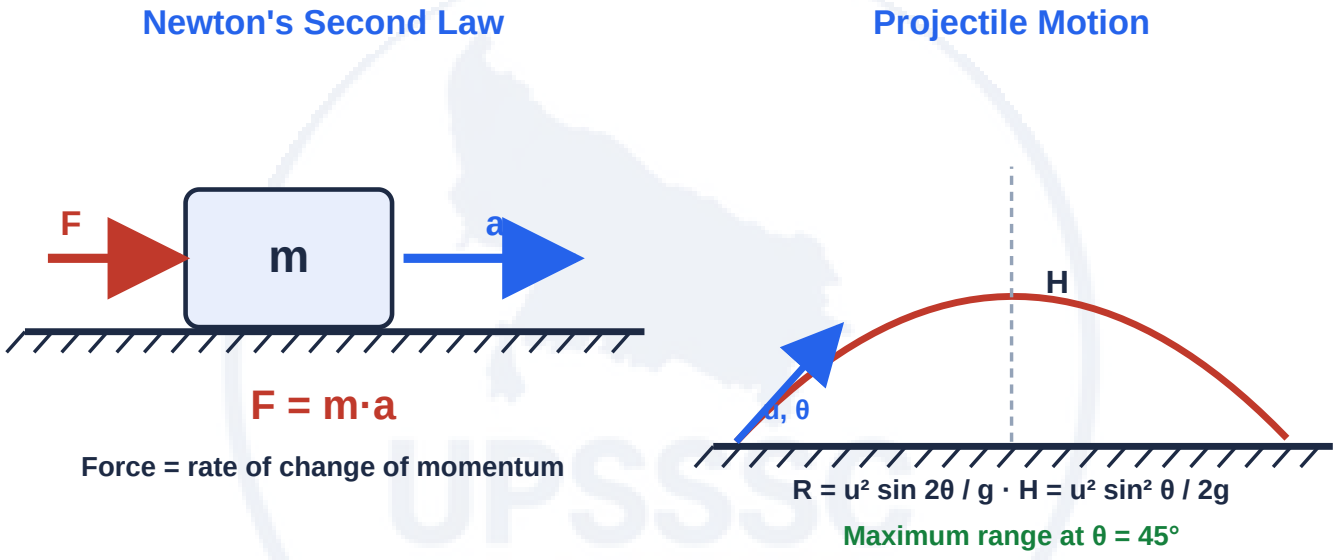


Fig 2.3: Newton's second law and path of a projectile

Derivation: Equations of motion under uniform acceleration

1. From definition a = (v − u)/t → **v = u + at**
2. Distance = average velocity × time = [(u + v)/2] t ; substituting v → **s = ut + ½at²**
3. Eliminating t → **v² = u² + 2as**
4. Distance in the n-th second: **s_n = u + a(2n − 1)/2**
5. **v–t graph**: slope = acceleration, area under it = displacement ; **s–t graph**: slope = velocity

Motion under gravity: a = g = 9.81 m/s² (downward). Body thrown up: maximum height H = u²/2g; time to top = u/g; total time of flight = 2u/g; returns with speed u. Body dropped from height h: v = √(2gh), t = √(2h/g).

Projectile motion: a body thrown with velocity u at angle θ — horizontal velocity u cos θ stays constant (no horizontal force), vertical motion is affected by g. **Time of flight T = 2u sin θ/g ; maximum height H = $u^2 \sin^2\theta/2g$; range R = $u^2 \sin 2\theta/g$**. Maximum range at **θ = 45°**: $R_{max} = u^2/g$. Complementary angles (30° and 60°) give equal range.

> **Example:** A car moving at 36 km/h accelerates at 2 m/s² for 5 s. Final velocity and distance?
> **Solution:**
> 1. u = 36 × 5/18 = 10 m/s (multiply by 5/18 to convert km/h to m/s)
> 2. v = 10 + 2 × 5 = **20 m/s** (72 km/h)
> 3. s = 10 × 5 + ½ × 2 × 25 = **75 m**

2.2 Newton's laws of motion — in depth

First law (law of inertia): a body at rest stays at rest and a body moving with uniform velocity keeps moving with it, unless compelled by an external unbalanced force to change that state. **Inertia** — the tendency not to change one's state; its measure is mass. When a bus starts suddenly passengers fall backward (the body wants to stay at rest), when it stops suddenly they fall forward.

Second law: the rate of change of momentum (p = mv) is proportional to the applied force and takes place in its direction: F = d(mv)/dt; for constant mass **F = ma**. This gives the definition and measure of force. **Impulse** = F × t = change of momentum — a cricketer pulls his hands back while catching to increase the time t and reduce the force F (less injury).

Third law: to every action there is an equal and opposite reaction — **but they act on two different bodies**, so they do not cancel each other. When a gun fires a bullet, the gun recoils; a rocket throws gases backward and moves forward itself; while walking we push the ground back and the ground pushes us forward.

Conservation of momentum: if no external force acts, the total momentum of a system remains constant: $m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$. Recoil velocity of a gun: V = mv/M.

Derivation: Apparent weight in a lift — a famous question

1. Two forces act on a person standing in a lift: weight mg (down) and floor reaction R (up) — R is the 'apparent weight' (what a scale shows)
2. **Accelerating up with a:** R − mg = ma → **R = m(g + a)** — feels heavier
3. **Accelerating down with a:** mg − R = ma → **R = m(g − a)** — feels lighter
4. **Free fall (cable breaks, a = g):** R = 0 — **weightlessness** ; uniform velocity (a = 0): R = mg

D'Alembert's principle: write F − ma = 0, i.e. add an imaginary **inertia force (−ma)** opposite to the acceleration — now a dynamic problem can be solved like static equilibrium (ΣF = 0). **Connected bodies (Atwood machine):** with $m_1 > m_2$ hanging over a pulley, $a = (m_1 - m_2)g/(m_1 + m_2)$ and string tension $T = 2m_1m_2g/(m_1 + m_2)$.

Example: A 60 kg person is in a lift accelerating upward at 2 m/s². Scale reading?
Solution:

1. R = m(g + a) = 60 × (9.81 + 2) = **708.6 N** (shows ≈ 72.2 kg)
2. Downward at 2 m/s²: R = 60 × 7.81 = **468.6 N** (≈ 47.8 kg)

2.3 Work, energy, power and circular motion

Work = force × displacement in the direction of force = F s cos θ, unit joule (J). Push a wall until you are tired, but if it doesn't move, in physics the **work is zero**! **Energy** = capacity to do work. **Kinetic energy** KE = ½mv²; **potential energy** PE = mgh. **Work-energy theorem:** net work done on a body = change in its kinetic energy. **Power** = rate of doing work = W/t = F × v, unit watt; 1 hp = 746 W.

Circular motion: a body moving in a circle at uniform speed keeps changing the direction of its velocity, so it has an acceleration — towards the centre, **centripetal acceleration a = v²/r = ω²r**. The force needed = **centripetal force F = mv²/r** (tension of a string with a stone, gravity on a satellite). **Centrifugal force** — an imaginary outward force felt by an observer rotating with the body.

Banking of roads: on a curve the outer edge of the road is raised so that the horizontal component of the normal reaction provides the centripetal force — **tan θ = v²/(rg)**. On railways raising the outer rail is called **super-elevation (cant)**: e = Gv²/(gr), G = gauge. On a flat unbanked road maximum safe speed v = √(μrg).

Example: Banking angle of a road for 54 km/h on a 50 m radius curve?
Solution:

1. v = 54 × 5/18 = 15 m/s
2. tan θ = 15²/(50 × 9.81) = 225/490.5 = 0.459
3. θ = **24.6°**

1. Key Points

- **Newton's first law (law of inertia):** a body continues in its state of rest or uniform motion unless an external force compels it to change. It gives the **definition of force** and of **inertia**.
- **Newton's second law:** the rate of change of momentum is proportional to the applied force and takes place in its direction: F = dp/dt = m·a. It gives the **measurement of force**.
- **Newton's third law:** to every action there is an equal and opposite reaction. Action and reaction act on different bodies, so they do not cancel.
- **Momentum:** p = m·v (kg·m/s). **Impulse:** J = F·t = Δp. If external force is zero, total momentum is conserved (rocket, recoil of gun).
- **Equations of motion (uniform acceleration):** v = u + at, s = ut + ½at², v² = u² + 2as. Distance in nth second s_n = u + a(2n − 1)/2.
- **D'Alembert's principle:** F − ma = 0; adding an inertia force (−ma) turns a dynamic problem into one of static equilibrium.

- **Work, power, energy:** work $W = F \cdot s \cdot \cos\theta$ (J); power $P = W/t = F \cdot v$ (W); kinetic energy $\frac{1}{2}mv^2$, potential energy mgh. Work–energy theorem: net work = change in kinetic energy.
- **Apparent weight in a lift:** accelerating up $R = m(g + a)$; accelerating down $R = m(g - a)$; free fall $R = 0$ (weightlessness).
- **Collision:** coefficient of restitution $e = (v_2 - v_1)/(u_1 - u_2)$. Perfectly elastic $e = 1$; perfectly inelastic $e = 0$ (bodies stick together).
- **Angular motion equations:** $\omega = \omega_0 + \alpha t$, $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$, $\omega^2 = \omega_0^2 + 2\alpha\theta$; and $v = r\omega$, $a_t = r\alpha$.
- **Newton's law in rotation:** $T = I\alpha$, where I is moment of inertia; rotational KE = $\frac{1}{2}I\omega^2$.
- **Conservation of energy:** with conservative forces, mechanical energy (KE + PE) remains constant; falling through h gives $v = \sqrt{(2gh)}$.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Second law	**$F = m \cdot a = dp/dt$**
Momentum / Impulse	**$p = mv$, $J = F \cdot t = \Delta p$**
Equations of motion	**$v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$**
Work	**$W = F\ s \cos\theta$**
Power	**$P = W/t = F \cdot v$**
Kinetic / Potential energy	**$\frac{1}{2}mv^2$, mgh**
Range of projectile	**$R = u^2 \sin 2\theta / g$ (max at $\theta = 45°$)**
Max height / Time of flight	**$H = u^2 \sin^2\theta / 2g$, $T = 2u \sin\theta / g$**
Circular motion	**$a = v^2/r$, $F = mv^2/r$**

3. Detailed Explanation

- **Inertia:** the property by which a body resists a change in its state. Mass is the measure of inertia — more mass, more inertia.
- **Examples of momentum conservation:** recoil of a gun when a bullet is fired: $m_g v_g = m_b v_b$; rocket propulsion works on the same principle.
- **Use of impulse:** pulling the hands back while catching a ball increases the time and reduces the force ($F = \Delta p/t$). Airbags and shock absorbers use the same idea.
- **Circular motion:** even at uniform speed in a circle there is a centripetal acceleration v^2/r towards the centre, requiring a centripetal force mv^2/r.
- **Projectile:** the horizontal velocity stays constant ($u \cos\theta$) and only g acts vertically. The range is the same for θ and $(90° - \theta)$.
- **Free fall:** with $u = 0$ and $a = g$: $v = gt$, $h = \frac{1}{2}gt^2$, $v^2 = 2gh$. Thrown up, max height $u^2/2g$ and total time $2u/g$.

- **Pulley (Atwood) system:** for masses $m_1 > m_2$, $a = (m_1 - m_2)g/(m_1 + m_2)$ and $T = 2m_1m_2g/(m_1 + m_2)$.
- **Vehicle on a curve:** on a level road max safe speed $v = \sqrt{(\mu gr)}$; banking angle $\tan\theta = v^2/(rg)$.

4. Comparison Table — Mass vs Weight

Basis	Mass	Weight
Definition	Quantity of matter	Earth's pull (mg)
Quantity	Scalar	Vector
Unit	kg	N
Depends on place	No	Yes (on g)

5. Solved Numerical Problems

Example 1. A 50 N force acts on a 10 kg body. Find the acceleration and the distance moved in 4 s from rest.

Solution:

1. $a = F/m = 50/10 =$ **5 m/s²**
2. $s = ½ \times 5 \times 4^2 =$ **40 m**

Example 2. A 20 g bullet leaves a 4 kg gun at 400 m/s. Find the recoil velocity of the gun.

Solution:

1. Momentum conservation: $4 \times v = 0.02 \times 400$
2. $v = 8/4 =$ **2 m/s**

Example 3. A 60 kg man stands in a lift accelerating up at 2 m/s². Find the floor reaction (g = 10).

Solution:

1. $R = m(g + a) = 60 \times 12 =$ **720 N**

Example 4. A projectile is thrown at 20 m/s at 45°. Find the range and maximum height (g = 10).

Solution:

1. $R = u^2 \sin 90°/g = 400/10 =$ **40 m**
2. $H = u^2 \sin^2 45°/2g = 400 \times 0.5/20 =$ **10 m**

Example 5. A stone is thrown vertically up at 20 m/s (g = 10). Find the maximum height and total time.

Solution:

1. $H = u^2/2g = 400/20 =$ **20 m**
2. $T = 2u/g =$ **4 s**

Example 6. Masses of 5 kg and 3 kg hang over a pulley (g = 10). Find the acceleration and tension.

Solution:

1. a = (5 − 3) × 10/8 = **2.5 m/s²**
2. T = 2 × 5 × 3 × 10/8 = **37.5 N**

6. Frequently Asked Facts

- Newton's first law defines force; the second law measures it.
- Action and reaction act on different bodies.
- Maximum range at 45°; then H = R/4.

7. Note (Confusing Facts)

- Unit of impulse N·s is the same as that of momentum kg·m/s.
- In free fall weight is not zero; the apparent weight is zero.
- At the top of a projectile's path the velocity is not zero but u cos θ.

Remember: One line: F = ma, p = mv, J = Ft = Δp. Assertion–Reason: (A) A gun recoils on firing. (R) Momentum is conserved. — Both true; R explains A.

30 Most Important One-Liner Facts — Laws of Motion

1. Law of inertia is?

— Newton's first law.

2. Measure of inertia?

— Mass.

3. Law for measuring force?

— Second law (F = ma).

4. Action–reaction law?

— Third law.

5. Formula of momentum?

— p = mv.

6. Impulse equals?

— Change in momentum.

7. Unit of impulse?

— N·s.

8. Rocket works on?

— Conservation of momentum.

9. Force added in D'Alembert's principle?

— Inertia force (−ma).

10. Formula of work?

— W = F s cos θ.

11. Formula of power?

— P = F·v.

12. 1 HP equals?

— 746 W.

13. Kinetic energy?

— ½mv².

14. Reaction in a lift accelerating up?

— m(g + a).

15. Apparent weight in free fall?

— Zero.

16. Angle for maximum range?

— 45°.

17. Range of projectile?

— $u^2 \sin 2\theta / g$.

18. Centripetal acceleration?

— v^2/r.

19. e for perfectly elastic collision?

— 1.

20. e for perfectly inelastic collision?

— 0.

21. Relation of v and ω?

— $v = r\omega$.

22. Newton's law in rotation?

— $T = I\alpha$.

23. Rotational kinetic energy?

— $\frac{1}{2}I\omega^2$.

24. Velocity after falling h?

— $\sqrt{(2gh)}$.

25. Max height when thrown up?

— $u^2/2g$.

26. Total time (vertical throw)?

— $2u/g$.

27. Atwood machine acceleration?

— $(m_1 - m_2)g/(m_1 + m_2)$.

28. Max speed on level curve?

— $\sqrt{(\mu gr)}$.

29. Banking angle?

— $\tan\theta = v^2/rg$.

30. Unit of work?

— Joule.

Practice MCQs — Laws of Motion

Q1. Newton's first law gives the definition of:

(a) Force
(b) Momentum
(c) Work
(d) Energy

✔ **Answer: (a) Force**

Explanation: The first law defines force and inertia.

Q2. Impulse equals:

(a) Change in momentum
(b) Force × distance
(c) Mass × acceleration
(d) Kinetic energy

✔ **Answer: (a) Change in momentum**

Explanation: $J = F \cdot t = \Delta p$.

Q3. Action and reaction act on:

(a) The same body
(b) Different bodies
(c) Sometimes same, sometimes different
(d) Moving bodies only

✔ **Answer: (b) Different bodies**

Explanation: That is why they do not cancel each other.

Q4. The range of a projectile is maximum when the angle of projection is:

(a) 30°
(b) 45°
(c) 60°
(d) 90°

✔ **Answer: (b) 45°**

Explanation: $\sin 2\theta = 1$ gives $\theta = 45°$.

Q5. Apparent weight of a man in a lift going down with acceleration a:

(a) m(g + a)
(b) m(g − a)
(c) mg
(d) Zero

✔ **Answer: (b) m(g − a)**

Explanation: R = m(g − a).

Q6. The principle that converts a dynamic problem into static equilibrium:

(a) D'Alembert's
(b) Lami's
(c) Pascal's
(d) Archimedes'

✔ **Answer: (a) D'Alembert's**

Explanation: An inertia force −ma is added.

Q7. 1 HP equals:

(a) 736 W
(b) 746 W
(c) 1000 W
(d) 550 W

✔ **Answer: (b) 746 W**

Explanation: 1 HP = 746 W (metric HP = 735.5 W).

Q8. Coefficient of restitution for a perfectly inelastic collision is:

(a) 0
(b) 0.5
(c) 1
(d) Infinity

✔ **Answer: (a) 0**

Explanation: Bodies stick together, e = 0.

Q9. A 5 kg body's velocity changes from 10 to 20 m/s in 2 s. The force is:

(a) 10 N
(b) 25 N
(c) 50 N
(d) 100 N

✔ **Answer: (b) 25 N**

Explanation: a = 10/2 = 5 m/s², F = 5 × 5 = 25 N.

Q10. Power is given by:

(a) F × s
(b) F × v
(c) m × v
(d) ½mv²

✔ **Answer: (b) F × v**

Explanation: P = W/t = F·s/t = F·v.

Q11. Velocity of a body falling freely through h:

(a) gh
(b) √(gh)
(c) √(2gh)
(d) 2gh

✔ **Answer: (c) √(2gh)**

Explanation: v² = 2gh.

Q12. Relation of torque and angular acceleration:

(a) $T = m\alpha$ (b) $T = I\alpha$

(c) $T = I\omega$ (d) $T = \frac{1}{2}I\omega^2$

✔ **Answer: (b) $T = I\alpha$**

Explanation: Rotational form of Newton's law.

Q13. Max height of a stone thrown up at 20 m/s (g = 10):

(a) 10 m (b) 20 m

(c) 40 m (d) 200 m

✔ **Answer: (b) 20 m**

Explanation: $u^2/2g = 20$.

Q14. Max safe speed of a vehicle on a level curve:

(a) $\sqrt{(\mu gr)}$ (b) μgr

(c) $\sqrt{(gr/\mu)}$ (d) gr

✔ **Answer: (a) $\sqrt{(\mu gr)}$**

Explanation: $\mu mg = mv^2/r$.

Q15. By the work–energy theorem, net work equals:

(a) Potential energy (b) Change in kinetic energy

(c) Momentum (d) Power

✔ **Answer: (b) Change in kinetic energy**

Explanation: $W = \Delta KE$.

SSC JE / UPSSSC JE Previous Year Questions — Laws of Motion

Q1. Newton's second law gives the measurement of:

SSC JE (Previous Year)

(a) Force (b) Inertia

(c) Energy (d) Work

✔ **Answer: (a) Force**

Explanation: Force is measured by F = ma.

Q2. D'Alembert's principle is based on:

SSC JE (Previous Year)

(a) Newton's first law (b) Newton's second law

(c) Newton's third law (d) Energy conservation

✔ **Answer: (b) Newton's second law**

Explanation: $F - ma = 0$ is a form of the second law.

Q3. Unit of momentum is: SSC JE (Previous Year)

(a) kg·m/s
(b) kg·m/s²
(c) N/s
(d) J·s

✔ **Answer: (a) kg·m/s**

Explanation: p = mv.

Q4. At the highest point of its path, the velocity of a projectile is: SSC JE (Previous Year)

(a) Zero
(b) u cos θ
(c) u sin θ
(d) u

✔ **Answer: (b) u cos θ**

Explanation: The vertical component is zero; the horizontal u cos θ remains.

Q5. Law of conservation of momentum follows from: SSC JE (Previous Year)

(a) First law
(b) Third law
(c) Hooke's law
(d) Pascal's law

✔ **Answer: (b) Third law**

Explanation: Action and reaction are equal and opposite.

Q6. SI unit of work: SSC JE (Previous Year)

(a) N
(b) W
(c) J
(d) N/m

✔ **Answer: (c) J**

Explanation: Joule = N·m.

UPSSSC ALERT

Chapter 3: Friction

Friction – Basic Concept

When one body slides or tends to slide over another, a force opposing the motion develops between the contact surfaces; this is called **friction**. It always acts tangential to the surfaces and opposite to the (tendency of) motion.

Understand in Detail — Friction

3.1 Friction — an 'enemy' and a 'friend'

Push a heavy box lying on the floor — at first it does not move at all, then it suddenly starts sliding, and once sliding, pushing becomes slightly easier. All this is due to **friction**: the force between two surfaces in contact that opposes their relative motion. Cause — no surface is perfectly smooth at the microscopic level; its bumps (asperities) interlock and stick to each other.

As a friend: without friction we cannot walk, car tyres cannot grip the road, brakes won't work, screws and nails won't hold, belt drives won't run. **As an enemy:** wasted energy in machines, heat, wear — hence lubrication and ball bearings.

Types: static — as long as the body has not moved; it is **self-adjusting** — it resists exactly as much as you push, up to a maximum called **limiting friction**. **Kinetic (dynamic)** — while sliding; slightly less than limiting (hence easier once sliding). **Rolling** — the least; that is why the wheel was a revolutionary invention. Order: **limiting static > kinetic > rolling**.

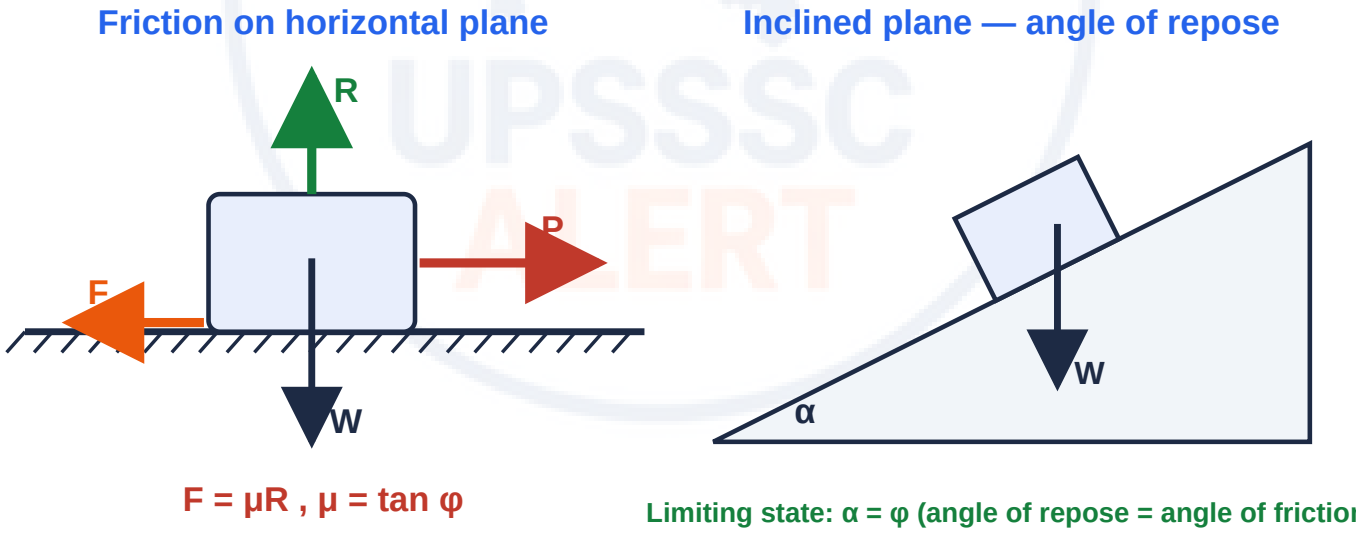


Fig 2.4: Force of friction and angle of repose on an inclined plane

3.2 Laws of friction, coefficient, angle of friction and angle of repose

Coulomb's laws of dry friction: (1) friction always opposes motion (or impending motion); (2) limiting friction is proportional to the normal reaction: **F = μ R**; (3) friction does not depend on contact area (a brick on edge or flat has the same friction); (4) depends on the nature of surfaces (material, roughness); (5) kinetic friction is nearly independent of velocity at low speeds.

Coefficient of friction μ = limiting friction/normal reaction — it has no unit; usually between 0 and 1 (but can exceed 1 for some surfaces).

Derivation: Angle of friction = angle of repose

1. **Angle of friction (φ):** the angle the resultant (S) of limiting friction F and normal reaction R makes with R: tan φ = F/R = **μ**
2. **Angle of repose (α):** the maximum inclination of a plane at which a body placed on it just begins to slide down under its own weight
3. On the incline: force down the plane = W sin α ; normal R = W cos α ; limiting friction = μ W cos α
4. On the point of sliding: W sin α = μ W cos α → **tan α = μ = tan φ → α = φ**
5. **Cone of friction:** the cone formed by revolving the resultant reaction at angle φ about the normal — if the applied force lies inside it, the body will not slide

Example: A 500 N box is on a horizontal floor, μ = 0.4. Minimum horizontal force to slide it? And if the force is applied at 30° upward?

Solution:

1. Horizontal: R = 500, P = μR = 0.4 × 500 = **200 N**
2. At 30° up: R = 500 − P sin30° ; P cos30° = 0.4(500 − 0.5P)
3. 0.866P + 0.2P = 200 → P = **187.6 N** — pulling slightly upward is easier because R (and friction) reduces
4. Minimum force occurs when the angle = angle of friction φ = $\tan^{-1}0.4$ = 21.8°: P_{min} = W sin φ = 500 × 0.371 = **185.7 N**

3.3 Motion on an inclined plane — up and down

A body (weight W) on an inclined plane (angle α) has three forces: weight W (vertically down), normal reaction R (perpendicular to the plane), friction F (opposite to impending motion, along the plane). Resolve weight into two components: **W sin α** down the plane and **W cos α** perpendicular to it.

Derivation: Force P needed parallel to the plane

1. R = W cos α ; F = μ W cos α
2. **Pulling up:** friction acts downward → **P = W(sin α + μ cos α)**
3. **Preventing sliding down** (or pulling down): friction acts upward → **P = W(sin α − μ cos α)**
4. Putting μ = tan φ: up P = W sin(α + φ)/cos φ ; down P = W sin(α − φ)/cos φ
5. If α < φ then W(sin α − μ cos α) is negative — the body will not slide even without support (**self-locking**)

Example: A 1000 N load on a 30° incline, μ = 0.2. Force parallel to the plane to (i) pull it up, (ii) prevent it sliding down?

Solution:

1. W sin30° = 500 N ; μ W cos30° = 0.2 × 1000 × 0.866 = 173.2 N
2. (i) P = 500 + 173.2 = **673.2 N**
3. (ii) P = 500 − 173.2 = **326.8 N**

3.4 Engineering applications of friction — ladder, screw jack, belt, clutch

Ladder problem: a ladder leaning on a wall has three forces — weight (at the middle), floor reaction (vertical R_1 + friction $\mu_1 R_1$ towards the wall), wall reaction (horizontal R_2 + friction $\mu_2 R_2$ upward). When the ladder is about to slip, ΣH = 0, ΣV = 0, ΣM = 0 give the minimum angle or how high a man can climb. For a smooth wall and rough floor: tan θ = 1/(2μ) (θ with the floor).

Screw jack: a screw is really an inclined plane wrapped round a cylinder; **helix angle α**: tan α = p/(πd) (p = pitch/lead). To lift load W the force at the screw circumference P = W tan(α + φ); force at the handle (length L) = P × (d/2)/L. **Efficiency η = tan α/tan(α + φ)**; maximum efficiency (1 − sin φ)/(1 + sin φ) when α = 45° − φ/2. A jack must be **self-locking** (the load must not come down by itself when the handle is released) — this needs α < φ, and then **η < 50%**.

Belt friction: the difference between the tight side (T_1) and slack side (T_2) of a belt wrapped on a pulley is possible only due to friction: $\mathbf{T_1/T_2 = e^{\mu\theta}}$ (θ = angle of wrap, in radians). That is why sailors hold a heavy ship by giving a rope a few turns round a bollard — every turn multiplies the tension ratio. **Power transmitted by a belt = $(T_1 - T_2)$ v**.

Clutches, brakes and bearings: a car's clutch (plate clutch) and brakes connect/stop power only through friction. Two theories of friction torque: **uniform pressure** (new clutches) — $T = (2/3)\mu W(R_1^3 - R_2^3)/(R_1^2 - R_2^2)$; **uniform wear** (old clutches, safer design) — $T = \mu W(R_1 + R_2)/2$. In bearings, friction circle radius = r sin φ ≈ μr.

Example: A belt with 180° wrap, μ = 0.3, slack-side tension 500 N. Tight-side tension? Power at 10 m/s belt speed?

Solution:

1. θ = π radian ; $e^{\mu\theta} = e^{0.3\pi} = e^{0.942} = 2.566$
2. $T_1 = 2.566 \times 500 =$ **1283 N**
3. $P = (1283 - 500) \times 10 = 7830$ W = **7.83 kW**

1. Key Points

- **Static friction:** friction while the body is at rest; it is self-adjusting and rises up to the limiting value $F_s = \mu_s R$.
- **Limiting friction:** the maximum friction force just before motion starts.
- **Kinetic (dynamic) friction:** friction during motion; it is less than limiting friction. Order: static (limiting) > sliding > rolling friction.
- **Coefficient of friction:** μ = F/R, where R is the normal reaction. It is dimensionless.
- **Angle of friction (φ):** the angle between the resultant reaction and the normal; tan φ = μ.
- **Angle of repose (α):** the maximum inclination of a plane at which a body just begins to slide under its own weight; α = φ.
- **Laws of dry (Coulomb) friction:** (i) F ∝ R; (ii) F is independent of contact area; (iii) F depends on the nature of surfaces; (iv) kinetic friction is nearly independent of speed at low speeds.
- **Types by lubrication:** dry, boundary (thin film) and fluid (viscous) friction.
- **Cone of friction:** the cone generated by revolving the resultant reaction about the normal; its semi-angle is φ.
- **Screw jack and friction:** a screw acts like an inclined plane; P = W tan(α + φ) and self-locking condition φ > α.
- **Belts and brakes:** in band brakes and belt drives $T_1/T_2 = e^{\mu\theta}$; θ in radians.

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Friction force	**$F = \mu R$**
Angle of friction	**$\tan\varphi = \mu$**
Angle of repose	**$\alpha = \varphi$**
Pull up the plane (parallel to plane)	**$P = W(\sin\alpha + \mu\cos\alpha)$**
Pull down the plane	**$P = W(\mu\cos\alpha - \sin\alpha)$**
Horizontal plane, pull at angle θ	**$P = \mu W / (\cos\theta + \mu\sin\theta)$**
Angle for least force	**$\theta = \varphi \rightarrow P_{min} = W\sin\varphi$**
Belt / rope friction	**$T_1/T_2 = e^{\mu\theta}$**

3. Detailed Explanation

- **Cause of friction:** interlocking of microscopic surface projections (asperities) and intermolecular adhesion.
- **Body on an inclined plane:** W sin α acts down the plane and friction μW cos α opposes it. If tan α ≤ μ, the body will not slide by itself.
- **Ladder problems:** taking normal reactions and friction at wall and floor, apply ΣFx = ΣFy = ΣM = 0.
- **Advantages and disadvantages:** walking, brakes, belt drives and the grip of nuts and bolts depend on friction; but it also causes energy loss, wear and heat. Lubrication and ball bearings reduce it.
- **Rolling friction:** the resistance due to deformation at contact during rolling; much smaller than sliding friction — hence wheels and bearings.
- **General ladder solution:** with a smooth wall there is only a normal reaction at the wall; at the floor R and μR. ΣFx: $\mu R_f = R_w$; ΣFy: $R_f = W$; ΣM about the foot gives the angle or μ. At limiting slip, tan θ = 1/(2μ) (smooth wall, weight at middle, θ with floor).
- **Force to prevent sliding down:** P = W(sin α − μ cos α), when tan α > μ.
- **Types of lubrication:** hydrodynamic (oil film from shaft motion), hydrostatic (pressure from an external pump) and boundary lubrication.

4. Comparison Table — Static vs Kinetic Friction

Basis	Static	Kinetic
State	At rest	In motion
Magnitude	0 to $\mu_s R$ (self-adjusting)	Nearly constant $\mu_k R$
Coefficient	μ_s (higher)	μ_k (lower)
Example	Car parked on a slope	Sliding box

5. Solved Numerical Problems

Example 1. A 200 N body rests on a horizontal plane, μ = 0.3. Find the horizontal force needed to move it.

Solution:

1. P = μW = 0.3 × 200 = **60 N**

Example 2. If μ = 0.5, find the angle of friction and angle of repose.

Solution:

1. $\varphi = \tan^{-1} 0.5 =$ **26.57°**
2. Angle of repose α = φ = **26.57°**

Example 3. A 500 N body lies on a 30° incline, μ = 0.2. Find the force parallel to the plane to move it up.

Solution:

1. P = W(sin 30° + μ cos 30°) = 500(0.5 + 0.2 × 0.866)
2. P = 500 × 0.673 = **336.6 N**

Example 4. A body is just on the point of sliding on a 25° incline. Find the coefficient of friction.

Solution:

1. μ = tan 25° = **0.466**

Example 5. A 1000 N body is on a 30° incline, μ = 0.2. Find the least force parallel to the plane to prevent it sliding down.

Solution:

1. P = W(sin 30° − μ cos 30°) = 1000(0.5 − 0.1732)
2. P = **326.8 N**

Example 6. In a band brake μ = 0.3, angle of contact 180°, T_2 = 200 N. Find T_1.

Solution:

1. $T_1 = 200 \times e^{0.3\pi} = 200 \times 2.566$
2. T_1 = **513 N**

6. Frequently Asked Facts

- Friction force does not depend on the area of contact.
- Angle of repose = angle of friction; μ = tan φ.
- Limiting friction > kinetic friction > rolling friction.

7. Note (Confusing Facts)

- Static friction is not always μR; μR is only its limiting (maximum) value.
- μ can exceed 1 (e.g. rubber on rubber).
- When pulling at an angle θ, the least force is needed when θ = φ.

Remember: One line: F = μR, tan φ = μ, α = φ. Assertion–Reason: (A) Moving a load on wheels is easier. (R) Rolling friction is less than sliding friction. — Both true; R explains A.

30 Most Important One-Liner Facts — Friction

1. Formula of friction force?

— F = μR.

2. Unit of coefficient of friction?

— Dimensionless.

3. Relation of φ and μ?

— tan φ = μ.

4. Angle of repose equals?

— Angle of friction.

5. Friction just before motion?

— Limiting friction.

6. Friction during motion?

— Kinetic friction.

7. Least friction?

— Rolling friction.

8. Does friction depend on area?

— No.

9. Direction of friction?

— Opposite to motion.

10. Self-adjusting friction?

— Static friction.

11. Effect of lubrication?

— Reduces friction.

12. Friction with thin lubricant film?

— Boundary friction.

13. Friction with thick film?

— Fluid (viscous) friction.

14. Force to pull up incline?

— $W(\sin\alpha + \mu\cos\alpha)$.

15. Angle of least pull?

— Angle of friction φ.

16. Belt tension ratio?

— $e^{\mu\theta}$.

17. Condition for no sliding?

— $\tan\alpha \le \mu$.

18. Cause of friction?

— Interlocking of surface asperities.

19. Laws of dry friction by?

— Coulomb.

20. Brakes work on?

— Friction.

21. Semi-angle of cone of friction?

— φ.

22. Self-locking condition of screw jack?

— $\varphi > \alpha$.

23. Force to prevent sliding down?

— $W(\sin\alpha - \mu\cos\alpha)$.

24. Band brake tension ratio?

— $e^{\mu\theta}$.

25. Needed for hydrodynamic lubrication?

— Shaft motion.

26. Hydrostatic lubrication?

— Pressure from external pump.

27. Equations in ladder problems?

— $\Sigma Fx = \Sigma Fy = \Sigma M = 0$.

28. Friction on smooth surface?

— Zero.

29. Energy from friction?

— Heat.

30. μ of rubber on rubber?

— Can exceed 1.

Practice MCQs — Friction

Q1. Friction force does not depend on:

(a) Normal reaction

(b) Nature of surfaces

(c) Area of contact

(d) Coefficient of friction

✔ **Answer: (c) Area of contact**

Explanation: Coulomb's law — independent of area.

Q2. Relation between angle of friction φ and μ:

(a) μ = sin φ
(b) μ = cos φ
(c) μ = tan φ
(d) μ = cot φ

✔ **Answer: (c) μ = tan φ**

Explanation: tan φ = F/R = μ.

Q3. The angle of repose is equal to:

(a) Angle of friction
(b) 90° − φ
(c) 2φ
(d) φ/2

✔ **Answer: (a) Angle of friction**

Explanation: In the limiting state α = φ.

Q4. The correct order is:

(a) Rolling > sliding > static
(b) Static > sliding > rolling
(c) Sliding > static > rolling
(d) All equal

✔ **Answer: (b) Static > sliding > rolling**

Explanation: Limiting/static friction is the greatest, rolling the least.

Q5. Weight 100 N, μ = 0.25 — horizontal force to move it:

(a) 25 N
(b) 40 N
(c) 75 N
(d) 400 N

✔ **Answer: (a) 25 N**

Explanation: F = 0.25 × 100 = 25 N.

Q6. The unit of coefficient of friction is:

(a) N
(b) N/m
(c) No unit
(d) N·m

✔ **Answer: (c) No unit**

Explanation: It is a ratio of two forces.

Q7. A body on an incline will not slide by itself if:

(a) tan α > μ
(b) tan α ≤ μ
(c) sin α > μ
(d) cos α < μ

✔ **Answer: (b) tan α ≤ μ**

Explanation: W sin α ≤ μW cos α gives tan α ≤ μ.

Q8. When pulling at an angle θ, the least force is needed when:

(a) θ = 0
(b) θ = 45°
(c) θ = φ
(d) θ = 90°

✔ **Answer: (c) θ = φ**

Explanation: P = μW/(cos θ + μ sin θ) is minimum at θ = φ.

Q9. Friction with a very thin lubricant film is called:

(a) Dry
(b) Boundary
(c) Viscous
(d) Rolling

✔ **Answer: (b) Boundary**

Explanation: Surfaces remain in partial contact.

Q10. If a body is just about to slide on a 30° incline, μ is:

(a) 0.5
(b) 0.577
(c) 0.866
(d) 1.732

✔ **Answer: (b) 0.577**

Explanation: μ = tan 30° = 0.577.

Q11. The semi-angle of the cone of friction is:

(a) Half the angle of repose
(b) Angle of friction φ
(c) 90°
(d) 45°

✔ **Answer: (b) Angle of friction φ**

Explanation: The cone is generated by φ.

Q12. A screw jack is self-locking if:

(a) $\varphi < \alpha$
(b) $\varphi = \alpha$
(c) $\varphi > \alpha$
(d) $\varphi = 0$

✔ **Answer: (c) $\varphi > \alpha$**

Explanation: The load does not come down by itself.

Q13. In a band brake T_1/T_2 =

(a) μθ
(b) $e^{\mu\theta}$
(c) $e^{\theta/\mu}$
(d) $1 + \mu\theta$

✔ **Answer: (b) $e^{\mu\theta}$**

Explanation: Belt friction.

Q14. In hydrodynamic lubrication, the oil film is formed by:

(a) External pump
(b) Shaft motion
(c) Gravity
(d) Magnet

✔ **Answer: (b) Shaft motion**

Explanation: The rotating shaft drags the oil.

Q15. Coefficient of friction of a smooth surface:

(a) 0
(b) 0.5
(c) 1
(d) Infinity

✔ **Answer: (a) 0**

Explanation: Ideal smooth surface.

SSC JE / UPSSSC JE Previous Year Questions — Friction

Q1. The ratio of limiting friction to normal reaction is: SSC JE (Previous Year)

(a) Coefficient of friction
(b) Angle of friction
(c) Angle of repose
(d) Efficiency

✔ **Answer: (a) Coefficient of friction**

Explanation: $\mu = F/R$.

Q2. If the angle of friction is φ, the angle of repose is: SSC JE (Previous Year)

(a) φ
(b) 2φ
(c) φ/2
(d) 90° − φ

✔ **Answer: (a) φ**

Explanation: $\alpha = \varphi$.

Q3. Kinetic friction compared to limiting friction is: SSC JE (Previous Year)

(a) More
(b) Less
(c) Equal
(d) Infinite

✔ **Answer: (b) Less**

Explanation: Friction drops once motion starts.

Q4. According to Coulomb's laws, friction force is: SSC JE (Previous Year)

(a) Proportional to area
(b) Proportional to normal reaction
(c) Proportional to velocity
(d) Independent of load

✔ **Answer: (b) Proportional to normal reaction**

Explanation: $F \propto R$.

Q5. Maximum inclination at which a body stays on a plane without sliding: SSC JE (Previous Year)

(a) Angle of friction
(b) Angle of repose
(c) Helix angle
(d) Pressure angle

✔ **Answer: (b) Angle of repose**

Explanation: Angle of repose.

Q6. Friction acting on a moving body: SSC JE (Previous Year)

(a) Static
(b) Limiting
(c) Kinetic
(d) Zero

✔ **Answer: (c) Kinetic**

Explanation: Kinetic friction.
